



**ALLIANCE
UNIVERSITY**
Private University established in Karnataka State by Act No. 34 of year 2010
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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

**Title: Importing business datasets from flat files & MySQL and
connecting with any other sources into Power BI**

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

GitHub Link: <https://github.com/PragnaChelsiya/powerbi-miniprojects>

Problem Statement

You are a Data Analyst in a retail company. The company maintains business data related to sales, customers, products, dates, and regions. Management requires a centralized reporting system to understand overall business performance. The objective is to import the required data into Power BI, clean and transform the data, establish relationships between the tables, and develop an interactive dashboard. The dashboard should help management analyze total sales, orders, customers, product-wise sales, regional performance, and monthly sales trends for effective data-driven decision-making.

Methodology / Steps Followed

Step 1: Data Collection

- Collected the required retail sales dataset containing sales, customer, product, date, category, and regional information.
- Identified the relevant tables and fields required for business analysis.
- Prepared the data for importing into Power BI.

Step 2: Data Import

- Opened **Power BI Desktop**.
- Imported the required dataset using the **Get Data** option.
- Loaded the required tables into Power BI for analysis.

Step 3: Data Cleaning and Transformation

- Removed duplicate records where required.
- Checked and handled missing or incorrect values.
- Renamed columns appropriately for easy understanding.
- Changed columns to the appropriate **data types**.
- Created and prepared the **Calendar/Date table** for monthly analysis.
- Prepared the Month field for analyzing monthly sales trends

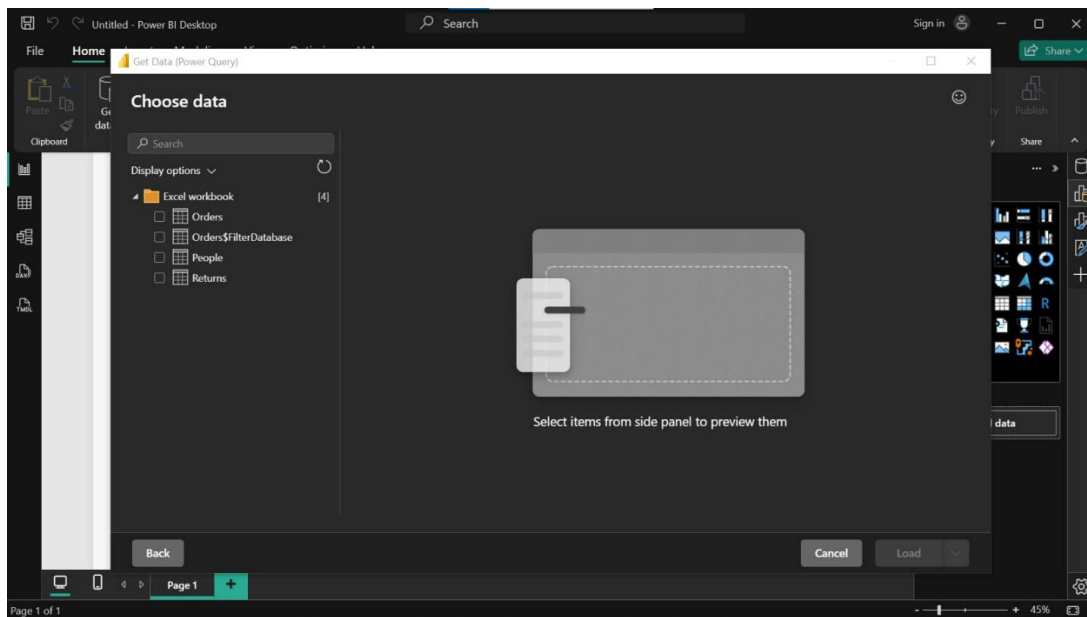
Step 4: Data Modeling

- Created relationships between the required tables.
- Established appropriate **one-to-many relationships** between related tables.
- Verified the relationships in **Model View**.
- Used the data model to enable filtering and interaction between different dashboard visuals.

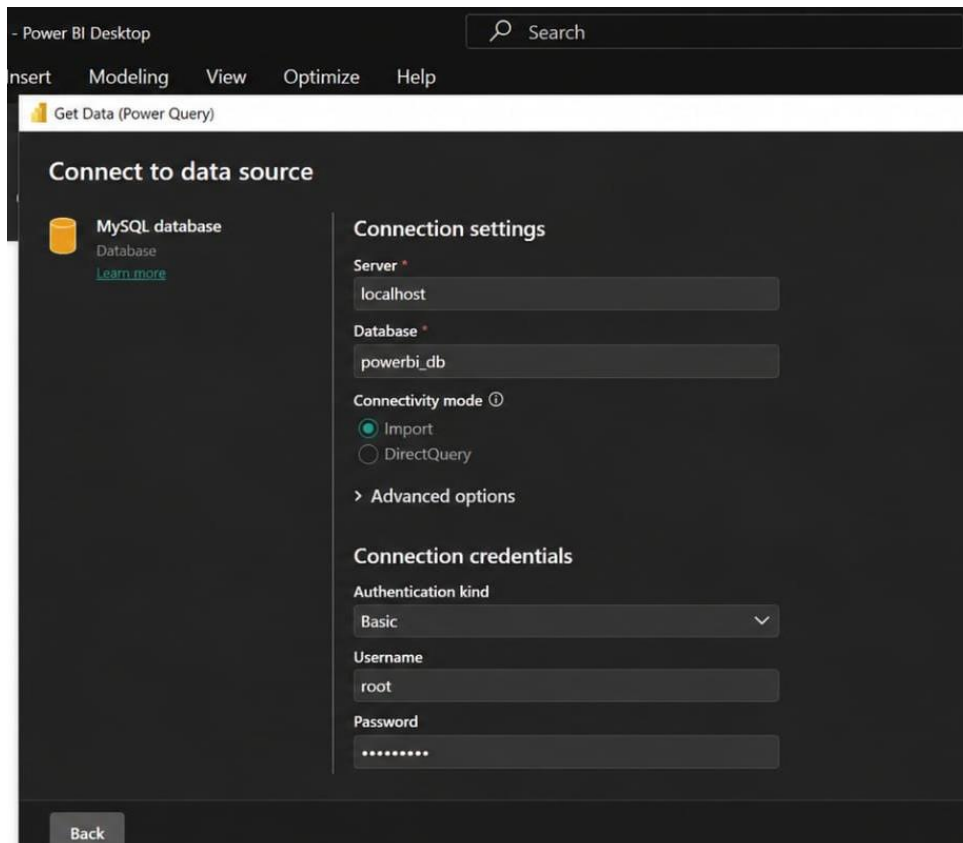
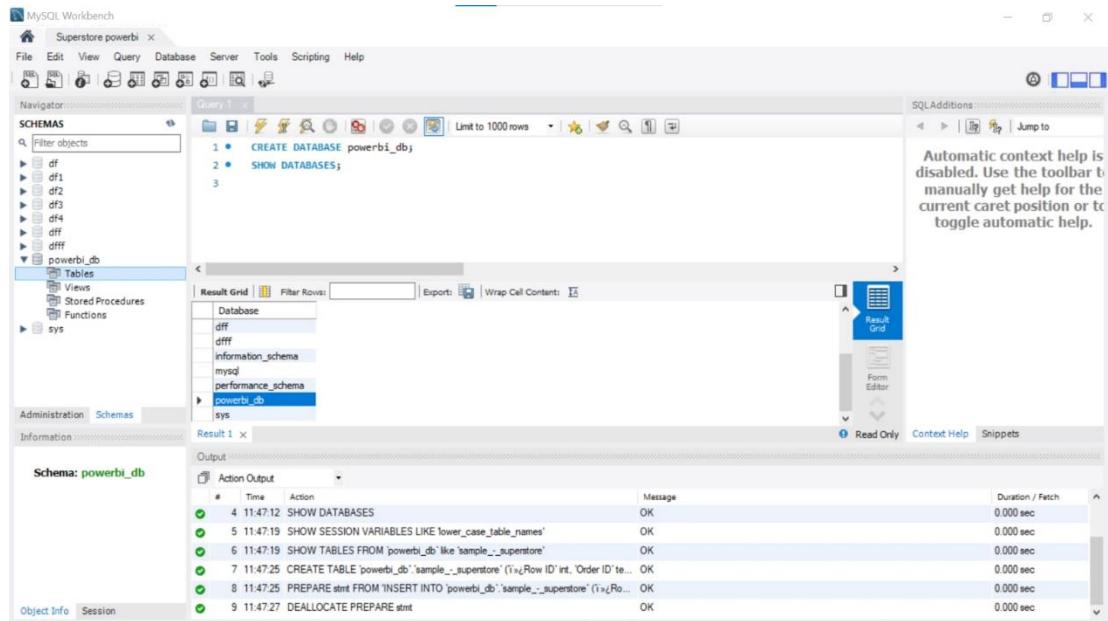
Step 5: Dashboard Development

- Created KPI cards for Total Sales, Orders, and Customers.
- Developed charts for Sales by Category and Region.
- Added slicers for Month, Category, and Region.
- Designed an interactive business dashboard.

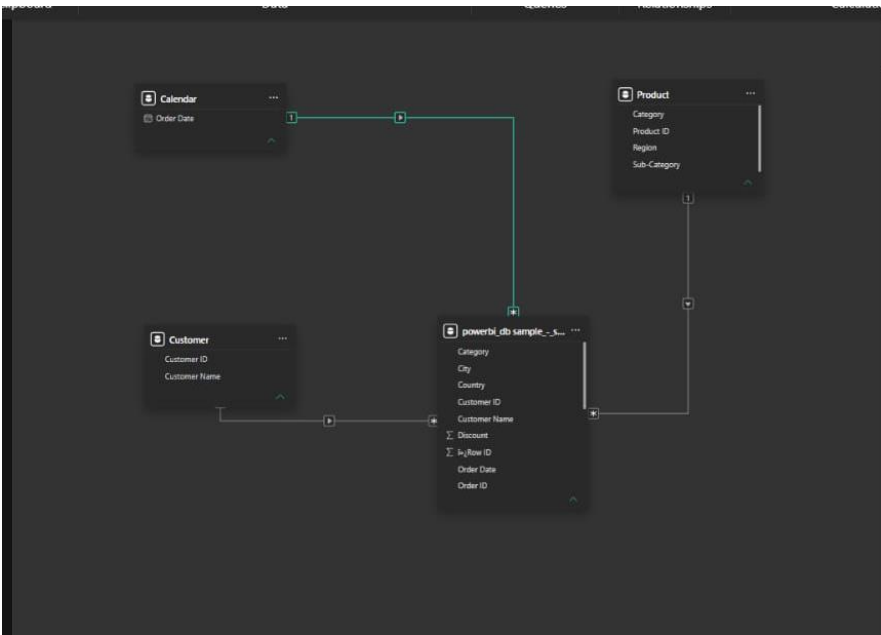
Output



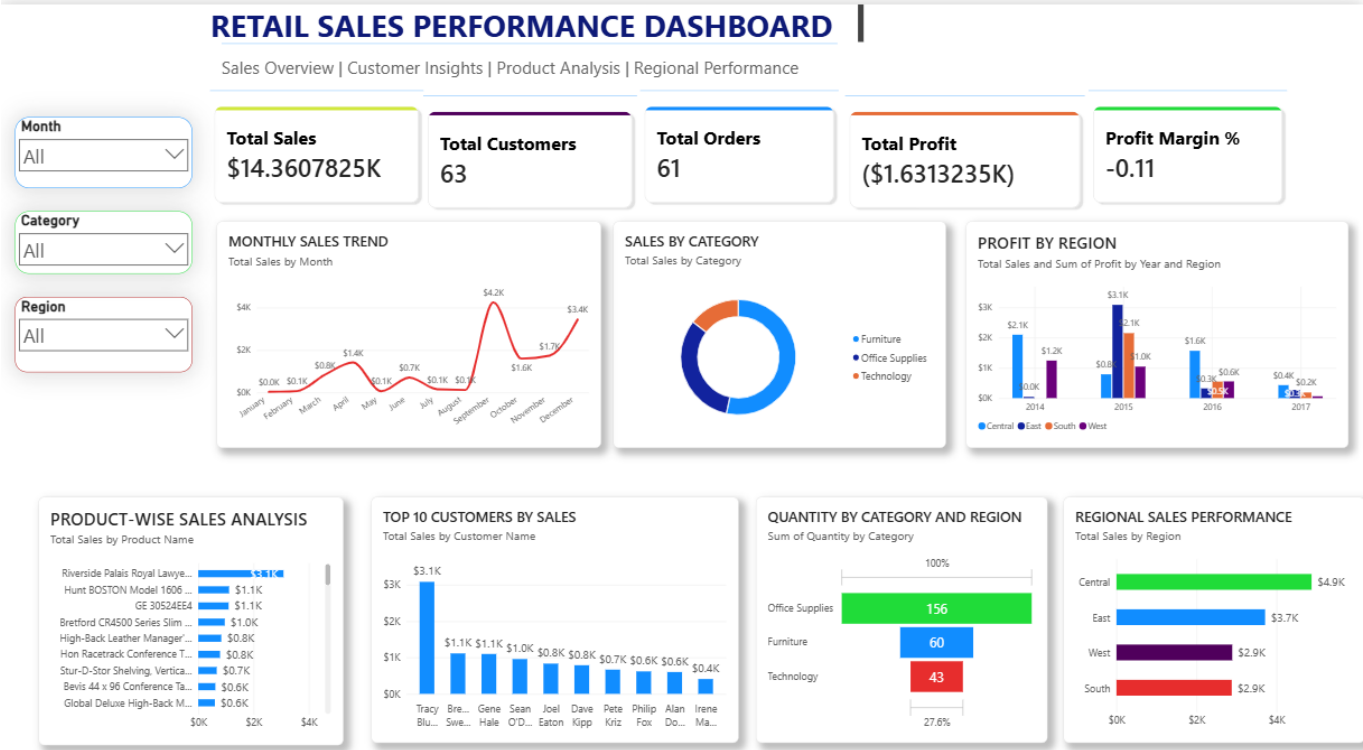
My Sql Connection:



Relationships:



Dashboard:



The final Power BI dashboard provides:

- Total Sales Performance
- Customer Analysis
- Product-wise Sales Analysis
- Regional Sales Performance
- Monthly Sales Trends
- Interactive Filters and Slicers

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Conclusion

This project successfully analyzed and transformed retail business data using Power BI to develop an interactive business dashboard. Data cleaning, transformation, data modeling, and table relationships were applied to prepare the data for accurate analysis. The developed dashboard provides a centralized view of sales performance, customer analysis, product-wise sales, regional performance, monthly sales trends, and other key business metrics. The interactive dashboard helps management understand business performance and supports effective data-driven decision-making.

Learning Outcomes

Through this project, I learned:

- How to import and prepare business data in **Power BI**.
- Data cleaning and transformation using **Power Query**.
- How to create and manage **relationships between multiple tables**.
- How to create **DAX measures and calculated columns** for analysis.
- How to create and format **KPI cards, charts, and interactive slicers**.
- How to design an **interactive business dashboard**.
- How to analyze **sales, customers, products, regions, and monthly trends**.
- How to generate **meaningful business insights from raw data**.